

Mixed Reality Guided Museum Tour: Digital Enhancement of Museum Experience

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Figure 1: Mixed Reality Guided Museum Tour in Mizuki Shigeru Museum

ABSTRACT

Museums as architectural structures rely on physical objects, which limit the content of their exhibits and the frequency of updates. The aim of the research was to create a richer museum visit experience by adding digital commentary and presentation to existing exhibits through a mixed reality-based guided tour system. We utilized headsets and visual positioning system to design contents that would enable a deeper understanding of the exhibits and an enjoyable experience. We found that more than 80 percent of participants to our experiment agreed with the effectiveness of the system and willingness to pay additional experience fees.

1 INTRODUCTION

Museums and art galleries are tourist attractions that satisfy intellectual curiosity. As the human desire for intellectual curiosity and tourism grows, so does the number of museums and the number of visitors. The exhibits and contents of museums are evolving year by year, and are not limited to the traditional exhibits of objects, but are also becoming more diverse, including exhibits of things and experiences. Some institutions are also providing guidance methods to deepen understanding of exhibits, such as human guides or audio guides. Existing work has introduced similar conceptual methods,

such as using a handheld device and markers [1] or using a headset in the limited space of a museum [2].

In this paper, we design and develop a tour guide system using mixed reality. Mixed reality is a technology that places digital content in a physical environment [3]. As a concrete demonstration experiment, we present our results in a museum that displays the life and work of a cartoonist. First, the requirements for the design of a mixed reality tour system are defined. Then, the technical configuration and content of the developed guide system are reviewed. Finally, the results of a user participation demonstration experiment conducted in a real museum in September 2022 will be reported. The usefulness and future potential of the guide system using MR technology will be discussed.

2 MR GUIDED MUSEUM TOUR

This paper proposes an exhibit enhancement and interpretation system using mixed reality to solve the architectural and physical limitations problems faced by museums. The system assumes that the visitor wears a transparent head-mounted display (HMD) on his head to tour the museum. By recognising the user's 3D position and the exhibits they are looking at, the transparent HMD presents 3D objects, 2D images, subtitled text commentary and audio guidance to the visitor. The aim is to enhance the museum experience by providing additional content that enhances the visitor's understanding of the exhibits, provides information tailored to the visitor's needs and provides commentary that is current and relevant.

There were two options for implementing MR in real space: a headset-based method or a smartphone-based method. We decided to use a headset-based MR method to prevent visitors from using their hands for this system. The headset used was Microsoft's HoloLens 2, a pair of mixed reality smart glasses developed and marketed by Microsoft that can be worn on the head to enable hands-free recognition of real space and 2D/3D objects in space.

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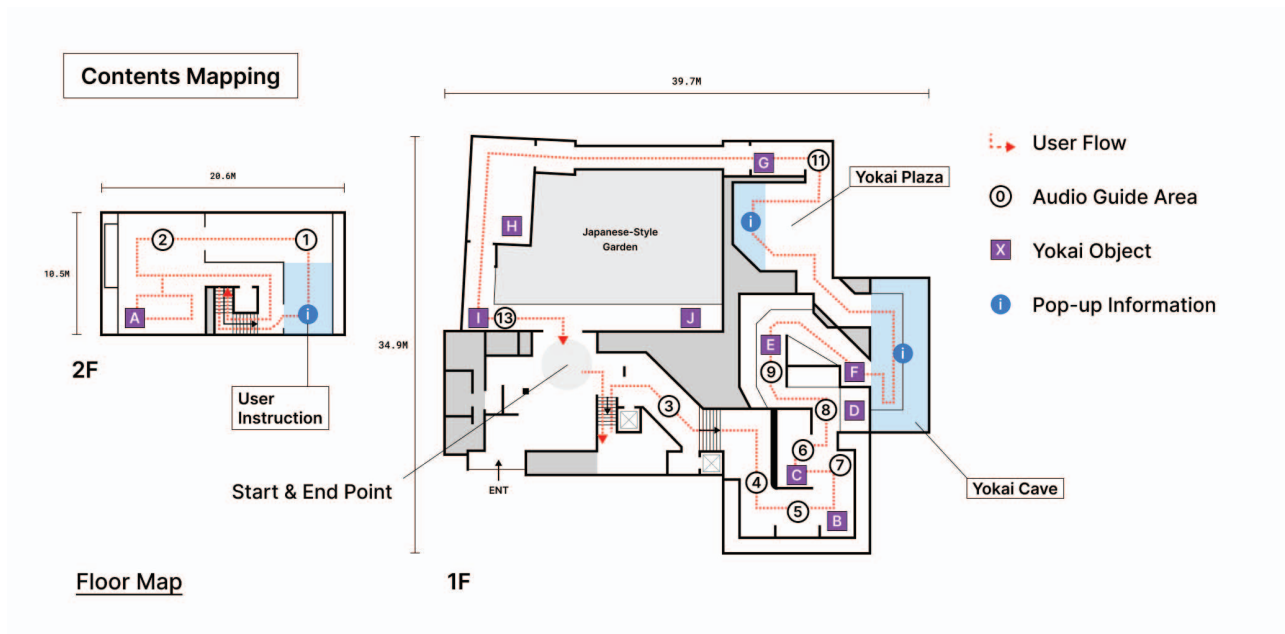


Figure 2: Floor Map and Digital Contents

The MR application is built using Unity, a game development engine, and reads locally stored digital content. The application sends camera data to a cloud computing platform for visual position estimation.

2.1 Visual Positioning System

The application running on HoloLens 2 used Immersal's VPS (Visual Positioning System) as its self-positioning estimation technology. This technology enables highly accurate recognition of the museum's location and exhibits. To place digital objects in different locations in the museum, a spatial map must be created using the Immersal application.

2.2 Area Segmentation for Occlusion and Memory Management

The application is divided into several areas within the museum and the content is loaded sequentially. The museum has many turns, and when visitors look back, they may see digital content that they have already passed, or that they will see in the future, through the walls. For this reason, the wall occlusion setting was also used. Given the limitations of the headset's CPU and memory, we had to manage the system's computing resources and content capacity in order to present audio, 2D images and 3D models while constantly estimating the user's own position as they moved around the museum. Once the user has passed, the application deletes the data it has passed in order to reduce memory consumption.

2.3 Mixed Reality Contents

We asked the Mizuki Shigeru Museum to work with us to create content and experiences that are appropriate for the museum. The Mizuki Shigeru Museum is a museum where visitors can experience the life and work of Japanese manga (cartoon) artist Shigeru Mizuki. Yokais are Japanese ghost-like creatures that Mr Mizuki wrote about a lot in his work. In this museum, audio guide, 2D and 3D Yokais and gates, text explanation of Yokais are digitally placed. Figure 2 shows the spatio-temporal arrangement of digital contents in the museum.

3 EXPERIMENT

An evaluation of the effectiveness of this system was conducted with 93 visitors to the Mizuki Shigeru Museum as subjects over a four-day period from 22-25 September 2022. The subjects were given an MR headset at the reception desk and a brief explanation of how to use the system, and then spent 40-60 minutes touring the museum. The 84 per cent of participants gave positive feedback and more than half of them said they would be willing to pay double the museum ticket price.

4 CONCLUSION

We proposed a mixed reality museum tour that can be used throughout the museum for about 45 minutes. The experiment showed that the participants evaluated their MR experience and answered that the explanations and effects shown in the headset were useful.

ACKNOWLEDGMENTS

The authors wish to thank Mizuki Production, Mizuki Museum, Microsoft Japan, and Immersal for their cooperation. This research result was obtained from the commissioned research (No.05401) by National Institute of Information and Communications Technology (NICT), Japan.

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